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I conducted a semi-supervised learning experiment using the UCI Optical Recognition of Handwritten Digits dataset, which contains images reduced to 8x8 pixels and represented as 64 numerical features. Each digit (0–9) serves as a class label, making this a 10-class classification problem. The goal of this experiment was to compare semi-supervised learning methods with a fully supervised baseline, analyze how performance changes as the proportion of labeled data varies, and identify strategies to improve future performance.

At low label proportions, semi-supervised methods performed better. Below 1%, the SVM failed to generalize. Label Propagation used data structure to spread labels and achieved the highest macro F1-scores. Self-Training improved over the baseline but was unstable. It sometimes generated noisy pseudo-labels that hurt accuracy. When label availability increased to 5–10%, the SVM improved. It matched or slightly outperformed the semi-supervised methods. This showed that semi-supervised learning helps when labels are scarce. Its advantage drops as more labels become available.

Label Propagation was more effective than Self-Training at low label proportions. It used geometric relationships to infer missing labels. Self-Training needed more labeled data to stabilize. At 1%, Label Propagation outperformed Self-Training. At 10%, both performed similarly. Performance varied by digit. Digit 6 had strong F1-scores even with few labels. Its cluster was compact and easy to separate. Label Propagation worked well in this case. Digit 8 performed poorly with few labels. Its cluster overlapped with digits 3 and 9. This caused errors in both propagation and pseudo-labeling. More labels improved boundary detection.

To improve future performance, use stratified sampling to ensure all classes are represented in the labeled subset. This prevents missing classes and avoids undefined metrics during evaluation. For Label Propagation, reduce the number of neighbors to five and limit the number of iterations to 300 with a tolerance of 1e-3. These changes improve stability and reduce computation time. For Self-Training, apply early stopping and set a confidence threshold of 0.75 to filter out low-confidence pseudo-labels. This helps maintain label quality and reduces noise. Replace the SVM with Logistic Regression to speed up training while maintaining similar accuracy. Finally, apply the manifold assumption. This assumes nearby samples share similar labels and improves generalization, especially for image-based datasets.

Overall, the results show that Label Propagation performs best when labeled data are scarce, Self-Training improves as labels increase, and the supervised SVM remains strongest when abundant labels are available. These insights highlight the value of semi-supervised methods in low-label scenarios and suggest practical strategies to optimize their performance.